MSG No. 164.90 $P_c\bar{3}m1$ [Type IV, trigonal]

Table 1: Wyckoff site: `2a`, site symmetry: `-3m`.

No.	position	mapping
1	$[0,\ 0,\ 0]$	`[1,2,3,4,5,6,7,8,9,10,11,12]`
2	$\left[0,\ 0,\ \frac{1}{2}\right]$	`[13,14,15,16,17,18,19,20,21,22,23,24]`

Table 2: Wyckoff site: `2b`, site symmetry: `-3'm`.

No.	position	mapping
1	$\left[0,\ 0,\ \frac{1}{4}\right]$	`[1,2,3,10,11,12,16,17,18,19,20,21]`
2	$\left[0,\ 0,\ \frac{3}{4}\right]$	`[4,5,6,7,8,9,13,14,15,22,23,24]`

Table 3: Wyckoff site: `4c`, site symmetry: `3m`.

No.	position	mapping
1	$[0,\ 0,\ z]$	`[1,2,3,10,11,12]`
2	$[0,\ 0,\ -z]$	`[4,5,6,7,8,9]`
3	$\left[0,\ 0,\ z+\frac{1}{2}\right]$	`[13,14,15,22,23,24]`
4	$\left[0,\ 0,\ \frac{1}{2}-z\right]$	`[16,17,18,19,20,21]`

Table 4: Wyckoff site: `4d`, site symmetry: `3m`.

No.	position	mapping
1	$\left[\frac{1}{3},\ \frac{2}{3},\ z\right]$	`[1,2,3,10,11,12]`
2	$\left[\frac{2}{3},\ \frac{1}{3},\ -z\right]$	`[4,5,6,7,8,9]`
3	$\left[\frac{1}{3},\ \frac{2}{3},\ z+\frac{1}{2}\right]$	`[13,14,15,22,23,24]`
4	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{1}{2}-z\right]$	`[16,17,18,19,20,21]`

Table 5: Wyckoff site: `6e`, site symmetry: `.2/m`.

No.	position	mapping
1	$\left[\frac{1}{2},\ 0,\ 0\right]$	`[1,4,7,10]`
2	$\left[0,\ \frac{1}{2},\ 0\right]$	`[2,5,8,11]`
3	$\left[\frac{1}{2},\ \frac{1}{2},\ 0\right]$	`[3,6,9,12]`
4	$\left[\frac{1}{2},\ 0,\ \frac{1}{2}\right]$	`[13,16,19,22]`
5	$\left[0,\ \frac{1}{2},\ \frac{1}{2}\right]$	`[14,17,20,23]`
6	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$	`[15,18,21,24]`

Table 6: Wyckoff site: `6f`, site symmetry: `.2'/m`.

No.	position	mapping
1	$\left[\frac{1}{2},\ 0,\ \frac{1}{4}\right]$	`[1,10,16,19]`

continued ...

Table 6

No.	position	mapping
2	$\left[0,\ \frac{1}{2},\ \frac{1}{4}\right]$	[2,11,17,20]
3	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{4}\right]$	[3,12,18,21]
4	$\left[\frac{1}{2},\ 0,\ \frac{3}{4}\right]$	[4,7,13,22]
5	$\left[0,\ \frac{1}{2},\ \frac{3}{4}\right]$	[5,8,14,23]
6	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{3}{4}\right]$	[6,9,15,24]

Table 7: Wyckoff site: 12g, site symmetry: .2.

No.	position	mapping
1	$[x,\ 0,\ 0]$	[1,4]
2	$[0,\ x,\ 0]$	[2,5]
3	$[-x,\ -x,\ 0]$	[3,6]
4	$[-x,\ 0,\ 0]$	[7,10]
5	$[0,\ -x,\ 0]$	[8,11]
6	$[x,\ x,\ 0]$	[9,12]
7	$\left[x,\ 0,\ \frac{1}{2}\right]$	[13,16]
8	$\left[0,\ x,\ \frac{1}{2}\right]$	[14,17]
9	$\left[-x,\ -x,\ \frac{1}{2}\right]$	[15,18]
10	$\left[-x,\ 0,\ \frac{1}{2}\right]$	[19,22]
11	$\left[0,\ -x,\ \frac{1}{2}\right]$	[20,23]
12	$\left[x,\ x,\ \frac{1}{2}\right]$	[21,24]

Table 8: Wyckoff site: 12h, site symmetry: .2'.

No.	position	mapping
1	$\left[x,\ 0,\ \frac{1}{4}\right]$	[1,16]
2	$\left[0,\ x,\ \frac{1}{4}\right]$	[2,17]
3	$\left[-x,\ -x,\ \frac{1}{4}\right]$	[3,18]
4	$\left[x,\ 0,\ \frac{3}{4}\right]$	[4,13]
5	$\left[0,\ x,\ \frac{3}{4}\right]$	[5,14]
6	$\left[-x,\ -x,\ \frac{3}{4}\right]$	[6,15]
7	$\left[-x,\ 0,\ \frac{3}{4}\right]$	[7,22]
8	$\left[0,\ -x,\ \frac{3}{4}\right]$	[8,23]
9	$\left[x,\ x,\ \frac{3}{4}\right]$	[9,24]
10	$\left[-x,\ 0,\ \frac{1}{4}\right]$	[10,19]
11	$\left[0,\ -x,\ \frac{1}{4}\right]$	[11,20]
12	$\left[x,\ x,\ \frac{1}{4}\right]$	[12,21]

Table 9: Wyckoff site: 12i, site symmetry: .m.

No.	position	mapping
1	$[x,\ -x,\ z]$	[1,11]
2	$[x,\ 2x,\ z]$	[2,12]

continued ...

Table 9

No.	position	mapping
3	$[-2x,\ -x,\ z]$	[3,10]
4	$[2x,\ x,\ -z]$	[4,9]
5	$[-x,\ x,\ -z]$	[5,7]
6	$[-x,\ -2x,\ -z]$	[6,8]
7	$\left[x,\ -x,\ z+\frac{1}{2}\right]$	[13,23]
8	$\left[x,\ 2x,\ z+\frac{1}{2}\right]$	[14,24]
9	$\left[-2x,\ -x,\ z+\frac{1}{2}\right]$	[15,22]
10	$\left[2x,\ x,\ \frac{1}{2}-z\right]$	[16,21]
11	$\left[-x,\ x,\ \frac{1}{2}-z\right]$	[17,19]
12	$\left[-x,\ -2x,\ \frac{1}{2}-z\right]$	[18,20]

Table 10: Wyckoff site: 24j, site symmetry: 1

No.	position	mapping
1	$[x,\ y,\ z]$	[1]
2	$[-y,\ x-y,\ z]$	[2]
3	$[-x+y,\ -x,\ z]$	[3]
4	$[x-y,\ -y,\ -z]$	[4]
5	$[y,\ x,\ -z]$	[5]
6	$[-x,\ -x+y,\ -z]$	[6]
7	$[-x,\ -y,\ -z]$	[7]
8	$[y,\ -x+y,\ -z]$	[8]
9	$[x-y,\ x,\ -z]$	[9]
10	$[-x+y,\ y,\ z]$	[10]
11	$[-y,\ -x,\ z]$	[11]
12	$[x,\ x-y,\ z]$	[12]
13	$\left[x,\ y,\ z+\frac{1}{2}\right]$	[13]
14	$\left[-y,\ x-y,\ z+\frac{1}{2}\right]$	[14]
15	$\left[-x+y,\ -x,\ z+\frac{1}{2}\right]$	[15]
16	$\left[x-y,\ -y,\ \frac{1}{2}-z\right]$	[16]
17	$\left[y,\ x,\ \frac{1}{2}-z\right]$	[17]
18	$\left[-x,\ -x+y,\ \frac{1}{2}-z\right]$	[18]
19	$\left[-x,\ -y,\ \frac{1}{2}-z\right]$	[19]
20	$\left[y,\ -x+y,\ \frac{1}{2}-z\right]$	[20]
21	$\left[x-y,\ x,\ \frac{1}{2}-z\right]$	[21]
22	$\left[-x+y,\ y,\ z+\frac{1}{2}\right]$	[22]
23	$\left[-y,\ -x,\ z+\frac{1}{2}\right]$	[23]
24	$\left[x,\ x-y,\ z+\frac{1}{2}\right]$	[24]